

12389371 Canada Ltd (Ez Voltage) – Breaker Panel Upgrade Training Course

Participants will gain practical knowledge and hands-on experience in upgrading residential electrical breaker panels, with a focus on safety, code compliance, and client communication. This training enhances their confidence and technical skills to deliver professional service in real-world residential settings.

COURSE OUTLINE

EZ Voltage – Breaker Panel Upgrade Training Course

Day 1 Summary (5 hours):

Day 1 introduces participants to residential electrical infrastructure, with a focus on understanding breaker panel design, common components, and typical household electrical layouts. It emphasizes the importance of safety by grounding participants in CSA and Ontario Electrical Safety Code standards. The day also builds client interaction skills through site assessments and communication strategies, preparing them to speak confidently with homeowners.

Unit 1 – Introduction to Residential Electrical Systems (1 Hour)

This unit introduces participants to the basic layout and functionality of residential electrical systems. Trainees explore the various types of breaker panels, including main service panels and subpanels, and learn how circuits are organized within a typical home layout. Emphasis is placed on understanding panel components such as breakers, busbars, and ground bars.

Unit 2 – Electrical Code & Safety Standards (1 Hour)

Participants learn about the Canadian Electrical Code (CEC) and Ontario Electrical Safety Code (OESC), focusing on sections that apply to residential installations and upgrades. This unit outlines the legal and safety requirements for installations, highlights recent updates, and covers how to interpret and apply these standards during on-site work.

Unit 3 – Site Assessments & Customer Communication (3 Hours)

This unit equips trainees with the skills to perform initial site assessments and accurately determine upgrade needs. Topics include identifying panel age and capacity limitations, evaluating wiring conditions, and communicating upgrade recommendations and safety concerns to clients. Emphasis is placed on building trust and clearly explaining the scope of work.

Activities and Exercises:

Trainees will identify breaker panel components using diagrams and mock panels,

engage in a code matching activity to understand ESA/OESC requirements, conduct a simulated site assessment with a panel condition checklist, and participate in a role-play exercise to practice client communication and quoting.

Day 2 Summary (7 hours):

Day 2 focuses on preparing for a panel upgrade through technical planning, regulatory navigation, and pricing strategies. Trainees will perform load calculations, explore ESA permitting processes, and gain insights into estimating costs. The day builds the foundation to confidently plan and quote residential upgrades.

Unit 4 – Load Calculations & Panel Sizing (2 Hours)

Participants will learn how to calculate electrical load requirements for residential homes using formulas and real-world scenarios. The unit explores factors like HVAC systems, major appliances, and future-proofing homes for additional loads (e.g., EV chargers or basement suites).

Unit 5 – Permit Process & Utility Coordination (1 Hour)

This unit covers how to apply for permits with the ESA, when utility coordination is required, and how to handle disconnections and inspections. Trainees will be introduced to relevant forms and ESA portal navigation.

Unit 6 – Pricing & Estimating for Panel Upgrades (4 Hours)

Participants will explore how to create accurate estimates for panel upgrades. Topics include cost breakdowns (materials, labour, permits, overhead), quoting strategies, and customer communication. Estimating tools and templates will be introduced.

Activities and Exercises:

Trainees will complete a load calculation worksheet using various appliance profiles, fill out a mock ESA permit application, and create a quote based on a sample client scenario with materials and labour included.

Day 3 Summary (8 hours):

This intensive day gives participants hands-on exposure to panel removal and installation. The emphasis is on real-world safety practices, proper mounting and wiring techniques, and code-compliant grounding and arc fault protection. Trainees will be involved in step-by-step physical demonstrations and simulations.

Unit 7 – Removal of Existing Panels (3 Hours)

Participants will observe and assist in removing outdated panels. Safety during disconnection, labeling wires, and preserving circuits will be emphasized.

Unit 8 – Installing New Panels (3 Hours)

Trainees will learn how to mount and wire a new panel, including torquing connections, bonding techniques, and managing cable entry and exits.

Unit 9 – Grounding and Arc Fault Protection (2 Hours)

This unit covers the principles and code requirements for grounding systems, bonding, and arc fault/ground fault protection.

Activities and Exercises:

Hands-on demonstrations using practice panels will be conducted, including full panel removal and replacement steps. Trainees will wire a mock panel, identify breakers, and simulate proper bonding and grounding techniques.

Day 4 Summary (5 hours):

Participants now focus on verifying installation integrity and resolving common issues. Troubleshooting skills are built through meter testing, scenario simulations, and final walkthroughs. The day also covers documentation and client education as part of the final sign-off process.

Unit 10 – Testing, Inspections & Troubleshooting (3 Hours)

This unit walks through how to test voltage, continuity, and breaker performance using multimeters. Trainees learn inspection steps and what red flags ESA may look for.

Unit 11 – Final Sign-Off & Client Handover (1 Hour)

Trainees learn to complete documentation (e.g., panel labels, ESA records) and educate clients on basic panel operation, breaker resetting, and maintenance.

Unit 12 – Panel Upgrade Case Studies (1 Hour)

A review of 2–3 real-world projects showing challenges (e.g., aluminum wiring, double taps) and how they were resolved.

Activities and Exercises:

Simulated faults will be introduced for trainees to troubleshoot using meters. Mock ESA inspection prep and client education walkthroughs will be conducted.

Day 5 Summary (5 hours):

Day 5 provides a full recap of the panel upgrade process, reinforcing key technical and safety lessons. Trainees participate in guided walkthroughs and role-play simulations to test their retention and readiness. The day ends with individual feedback and certification.

Unit 13 – Step-by-Step Review of Panel Upgrade Procedures (2 Hours)

A guided review of all major procedures covered throughout the week, reinforcing best practices and safety checks.

Unit 14 – Practical Walkthrough with Live Scenarios (2 Hours)

Trainees work through a full upgrade scenario using mock panels and materials, with supervision and evaluation by the instructor.

Unit 15 – Q&A and Individual Feedback Session (1 Hour)

Each trainee has a one-on-one with the instructor to review strengths, areas for improvement, and their readiness to perform work independently.

Activities and Exercises:

Mock upgrade walkthroughs, installation debrief, and peer evaluation. Group discussion and final Q&A close the session.

Activities:

Combined theory/practical review, group discussion

Fee per Participant: \$1,995

Certification Awarded: Residential Breaker Panel Upgrade Specialist